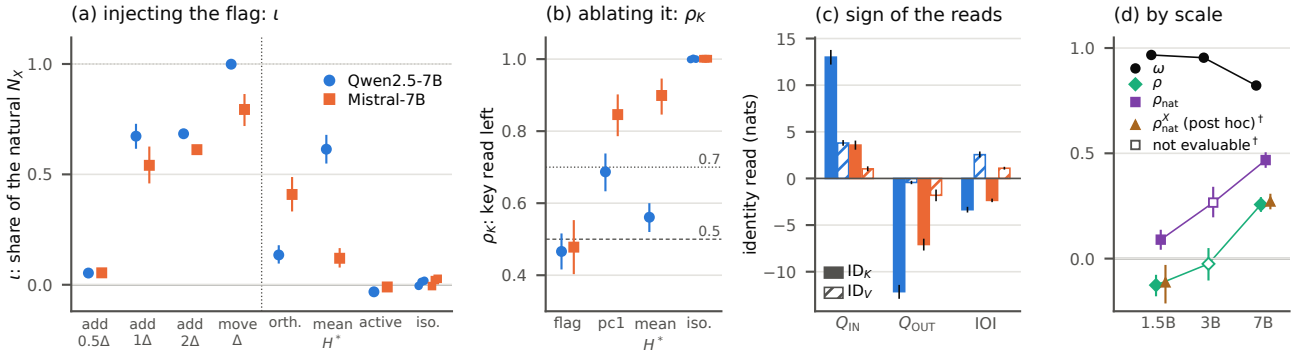
(a) injecting the flag: ι
ι: share of the natural N_X
1.0
0.5
0.0
Qwen2.5-7B
Mistral-7B
add 0.5Δ
add 1Δ
add 2Δ
move Δ
orth.
mean
active
iso.
H^*
(b) ablating it: ρ_K
ρ_K: key read left
1.0
0.8
0.6
0.4
0.7
0.5
flag
pc1
mean
iso.
H^*
(c) sign of the reads
identity read (nats)
15
10
5
0
−5
−10
ID_K
ID_V
Q_{IN}
Q_{OUT}
IOI
(d) by scale
1.0
0.5
0.0
ω
ρ
ρ_{nat}
ρ^X_{nat} (post hoc)†
not evaluable†
1.5B
3B
7B